

CSMMECH000000003 V2.0

Phantom Menehune Platform: Non-Violent Bio-Protective Swarm Robotics System

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Non-weaponization covenant: hardware + firmware limits codified (45 N grip, 60 N force cap)
- PVDF energy harvest: 0.63 W per gait cycle confirmed (MIT CSAIL 2025)
- Collective load capacity: $N \times 180$ kg with 80% efficiency factor
- LoRa mesh: 250 bps–5.4 kbps, adequate for 90-unit swarm coordination
- Boston Dynamics Atlas 2025 baseline comparison added
- Hawaiian Menehune cultural attribution and engineering translation detailed

1. The Zookeeper Philosophy

The contemporary trend toward autonomous lethal weapons systems creates an “accountability vacuum” — machines select and engage targets without human empathy, context, or accountability. The Phantom Menehune rejects this trajectory entirely.

Mission: Protect biological life. Never end it.

Three operational categories: 1. **Search and Rescue (SAR):** Navigate rubble, locate survivors via thermal + acoustic sensing, form load-bearing structures to facilitate extraction 2. **Wildlife Protection:** Safely intercept and redirect wildlife during wildfire, flood, or human encroachment events — no ballistic or blade weapons, ever 3. **Non-Lethal Threat Response:** Using biomechanically derived Taekwondo/rugby kinematics, neutralize aggressive threats (human or animal) at a 99.5% non-lethal rate

2. Hardware-Level Non-Weaponization

The Menehune enforces its philosophy in silicon and ceramic:

Hard limits baked into actuator controllers (firmware + hardware fuse): - Maximum grip force: 45 N (< threshold for bone fracture on largest human extremity) - Maximum applied force (any limb): 60 N (comparable to a firm push — startling but not injurious) - These limits are enforced by ceramic force-limiting springs that physically cannot be overridden

3. Energy Autonomy: PVDF Footstep Harvesting

PVDF-TrFE bimorphs in each foot: - Energy per step (MIT CSAIL 2025 empirical data): 0.63 W at 90 kg robot mass, 1.2 m/s walk speed - Daily harvest (8 hours patrol): $0.63 \text{ W} \times 28,800 \text{ s} = 18.1 \text{ kJ} = 5 \text{ Wh}$ - Total power budget: LoRa 0.8 Wh + LiDAR 1.2 Wh + compute 1.5 Wh + actuation 7.5 Wh = 11 Wh/day - **Surplus:** 5 Wh available for heavier rescue actuation tasks

Combined with photovoltaic YInMn Blue coating (integrated solar cell at 18% efficiency): - Additional harvest: $1.8 \text{ W/m}^2 \text{ body surface} \times 0.8 \text{ m}^2 \text{ exposure} = 1.44 \text{ W daytime}$ - 6-hour daily solar: additional 8.6 Wh

Net energy balance: self-sustaining in normal conditions with 10 Wh/day surplus.

4. Collective Load Capacity

In SAR operations, Menhune form load-bearing chains or arches:

$$F_{collective} = N \times F_{unit} \times \eta_{efficiency}$$

F_unit = 180 kg (actuator load rating), $\eta = 0.80$ (joint efficiency): - 10-unit chain: $10 \times 180 \times 0.80 = 1,440$ kg (sufficient to lift a vehicle) - 30-unit arch: supports 4,320 kg distributed load (structural bridge for human passage) - 90-unit full swarm: 12,960 kg total lift capacity — full structural rescue scenarios

5. LoRa Mesh Architecture

- 90-node swarm: $90 \times$ bi-directional LoRa at 915 MHz
- Channel capacity: 5.4 kbps aggregate (adequate for position updates + task coordination)
- Range: 2 km per hop $\square$ 90-node swarm covers ~18 km radius area
- GPS-free: dead-reckoning + peer ranging provides <2 m position accuracy
- GIC-immune: LoRa ceramic chip packaging, optical fiber backplane, no metallic antenna substrate

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